

✓ Single Neuron in Hidden Layer With Linear Activation

```
import pandas as pd
import tensorflow as tf

# Read CSV
data = pd.read_csv("student_data.csv")

# Input Features
X = data[['StudyHours', 'SleepHours', 'PlayingHours']]

# Output
y = data['Pass']

# Create Model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(
        units=1,
        activation='linear',
        input_shape=(3,)
    )
])

# Forward Pass
prediction = model(X)

print("Predictions")
print(prediction.numpy())

# Display Weights and Bias
weights, bias = model.layers[0].get_weights()

print("\nWeights")
print(weights)

print("\nBias")
print(bias)

# Model Summary
print("\nModel Summary")
model.summary()
```

```

Predictions
[[16.402784 ]
 [13.764487 ]
 [13.900768 ]
 [11.26247  ]
 [ 8.234019 ]
 [ 6.7879343]
 [ 4.149637 ]
 [ 2.7035522]
 [18.650928 ]
 [16.539066 ]
 [10.736036 ]
 [14.820418 ]
 [12.844837 ]
 [ 8.370301 ]
 [ 5.732003 ]
 [17.458715 ]
 [17.204844 ]
 [ 9.816386 ]
 [ 9.28995  ]
 [14.956699 ]]

```

```

Weights
[[ 1.0559312]
 [ 1.1922127]
 [-0.3901537]]

```

```

Bias
[0.]

```

Model Summary

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input\_shape`/`input\_dim`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 1)	4

Total params: 4 (16.00 B)

Trainable params: 4 (16.00 B)

Non-trainable params: 0 (0.00 B)

Single Neuron in Hidden Layer With Sigmoid Activation

```

import pandas as pd
import tensorflow as tf

# Read CSV
data = pd.read_csv("student_data.csv")

# Input Features
X = data[['StudyHours', 'SleepHours', 'PlayingHours']]

# Output
y = data['Pass']

# Create Model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(
        units=1,
        activation='sigmoid',
        input_shape=(3,)
    )
])

# Forward Pass
prediction = model(X)

print("Predictions")
print(prediction.numpy())

# Display Weights and Bias
weights, bias = model.layers[0].get_weights()

print("\nWeights")
print(weights)

```

```
print("\nBias")
print(bias)

# Model Summary
print("\nModel Summary")
model.summary()
```

```
Predictions
[[0.9978301 ]
 [0.99839884]
 [0.99957144]
 [0.9996839 ]
 [0.99991983]
 [0.99997914]
 [0.9999846 ]
 [0.999996 ]
 [0.99898714]
 [0.999419 ]
 [0.9995935 ]
 [0.9978866 ]
 [0.99967545]
 [0.99997854]
 [0.9999842 ]
 [0.99713635]
 [0.9997361 ]
 [0.9999177 ]
 [0.99989414]
 [0.9994341 ]]
```

```
Weights
[[-0.27809888]
 [ 1.0411559 ]
 [ 1.0675908 ]]
```

```
Bias
[0.]
```

```
Model Summary
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 1)	4

```
Total params: 4 (16.00 B)
Trainable params: 4 (16.00 B)
Non-trainable params: 0 (0.00 B)
```

Multi Layer Perceptron (Multiple Neurons in Hidden Layer)

```
import pandas as pd
import tensorflow as tf

# Read CSV
data = pd.read_csv("student_data.csv")

# Input Features
X = data[['StudyHours', 'SleepHours', 'PlayingHours']]

# Output
y = data['Pass']

# Create Model
model = tf.keras.Sequential([

    tf.keras.layers.Dense(
        units=4,
        activation='sigmoid',
        input_shape=(3,)
    ),

    tf.keras.layers.Dense(
        units=1,
        activation='sigmoid'
    )
])
```

```
])

# Forward Pass
prediction = model(X)

print("Predictions")
print(prediction.numpy())

# Hidden Layer Weights
weights1, bias1 = model.layers[0].get_weights()

print("\nHidden Layer Weights")
print(weights1)

print("\nHidden Layer Bias")
print(bias1)

# Output Layer Weights
weights2, bias2 = model.layers[1].get_weights()

print("\nOutput Layer Weights")
print(weights2)

print("\nOutput Layer Bias")
print(bias2)

# Model Summary
print("\nModel Summary")
model.summary()
```

Predictions

```
[[0.6277562 ]
 [0.61916983]
 [0.60719925]
 [0.6005187 ]
 [0.5958323 ]
 [0.593142 ]
 [0.59428114]
 [0.59146416]
 [0.6291039 ]
 [0.6163157 ]
 [0.6044547 ]
 [0.6254468 ]
 [0.5999644 ]
 [0.5887386 ]
 [0.58788854]
 [0.63251513]
 [0.6158175 ]
 [0.59367275]
 [0.59799784]
 [0.6140504 ]]
```

Hidden Layer Weights

```
[[ 0.86947274 -0.35800463 -0.53799033 -0.45042673]
 [-0.23711878 -0.6901007  0.39595187 -0.29503554]
 [ 0.27651381 -0.17371732  0.6478945  -0.10043496]]
```

Hidden Layer Bias

```
[0. 0. 0. 0.]
```

Output Layer Weights

```
[[ 0.5820999 ]
 [-0.50205934]
 [-0.20298165]
 [ 0.7149215 ]]
```

Output Layer Bias

```
[0.]
```

Model Summary

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_2"
```

Layer (type)	Output Shape	Param #